
UNIT 3 MODELS OF SCIENTIFIC EXPLANATION

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3.0 OBJECTIVES

After going through this unit, you will be able to:

- Explain the basic rules of formal reasoning;
- State the role of logical reasoning in the formulation of a research problem;
- Describe the basic rules of structural relationship between assumptions, theory and conclusions; and
- Appreciate the problems involved in systematic explanation of phenomena.

3.1 INTRODUCTION

Explanation has three meanings: **One** interpretation of explanation is to remove a perplexity or solve mystery i.e., 'puzzle solving'. The **second** meaning is to change the unknown into the known. The **third** interpretation of explanation is to give causes of phenomenon to be explained. The first two meanings are too relativistic and subjective. It is the third one emphasizing causal explanation

subserving regularity, and law like explanations that fits the meaning of scientific explanation. Here, we are concerned with one mainstream approach to science viz., Positivism. We may begin this unit by recognizing the complexity of explaining subject matter of philosophy of science. The first two units make it clear, that there is no general agreement on Positivism as the only approach to pursue knowledge. It is at least over fifty years, since when various limitations and misconception of the rules and propositions of Positivism have been brought to light. But yet, paradoxically, the mainstream notion of science and scientific explanation is dominated by the **tenets** of Positivism. This is partly because of the rigorous and prescriptive nature of positivist propositions which remain simple and attractive, and partly because of the open ended and speculative nature of the alternative approaches. The enthusiasm with which many mainstream economists still prefer to refer to Economics as a Positive Science is yet another indication that alternative approaches in the Philosophy of Science have not yet percolated to Economics. Because of these reasons this unit on 'Models of Scientific Explanation', and the next one on 'Debates on the Models of Scientific Explanation in Economics' are essentially confined to Positivist Models of Explanation.

Let us begin with a broad unified view of Positivism in terms of certain rules. It will be followed by certain rules of deduction or formal logical reasoning that underlie Positivist Models of explanation. Afterwards, the various Models of Explanation and their limitations, will be discussed.

3.2 UNIFIED VIEW OF RULES OF POSITIVISM

Our emphasis is on the Models of Scientific Explanation under Positivism. Positivism is procrustean in nature. Its propositions, often contested, evolved over a long period. It passed through several stages, from radical empiricism to logical positivism to logical empiricism. And the propositions of Positivism at various stages varied. Nevertheless, we shall confine here to a unified view of Positivism.

Kilakowski provides such a unified view by simplifying Positivism into a set of four rules:

K₁ The Rule of Phenomenalism

The Rule emphasizes phenomena as the basis of knowledge. Sensory experience is the basis. The basis of knowledge is only the record of that which is actually manifested in experience. It is phenomenon not noumina. It is existence, not essence. Simply, it is facts and objective facts only that form the basis of knowledge.

K₂ The Rule of Nouminalism

Nouminalism refers to Insight representative of facts. Any insight formulated in general terms can not have any real reference other than individual facts. Therefore every abstract science is abridging the recording of experience, and gives no extra independent knowledge.

K₃ The Rule of Value-free Statements

Knowledge is value-free. The rule refutes to call value judgments and normative statements as knowledge. There is no room for good or bad or ethical judgments in statement of science or knowledge.

K₄ The Rule of Unity of Science

There is only one method of scientific knowledge. The Unity of science refers to a single fundamental form from which all other laws are ultimately derived.

3.3 SEARCH FOR THE CRITERION OF COGNITIVE SIGNIFICANCE

In the above summation of Positivism into a set of rules, K₁, Rule of Phenomenalism, may appear to be closer to the early inductivist phase of Positivism. The inductivist view of science emphasized observation of facts and to proceed towards by inductive inference to the formulation of universal laws about these facts. But Positivism underwent considerable sophistication to accommodate deductive reasoning as a part of scientific explanation.

Yet another aspect of Positivism which is important in understanding its approach to scientific explanation is the criterion of cognitive significance. According to Positivism, the basic criterion for distinguishing ‘scientific knowledge’ or meaningful statements from ‘non-scientific’ or meaningless statements is **testability**. In the early phase the emphasis was on of **verification** – only complete verification by observational evidence to be considered empirically meaningful. But soon it was found to be too strict, and also faced the problem ‘induction’. Karl Popper suggested **falsification** testing instead of verification. But both verification and falsification tests are found to be too strict criteria, and alternatively, Rudolf Camap suggested **confirmation**. Confirmation is a relative concept. Instead of a single test for rejection or acceptance of a hypothesis, a series of positive results are supposed to increase the confirmation of it. Test instances confirmed or disconfirmed hypotheses, and hypotheses could be ranked according to their degree of confirmation relative to the available evidence. Later, emphasis shifted from testing individual theories to evaluation of competing theories on the basis of their relative degree of confirmation. But facts and testing remain basic ingredients of Positivist explanation.

3.4 RULES OF LOGIC OR RULES OF CORRECT REASONING

Just as facts and testing, rules of logic have become important part of the whole structure of scientific explanation under Positivism. Here we briefly review the main stay of deductive reasoning viz. ‘syllogism’ or ‘rules of logic’ or ‘rules of correct reasoning’.

3.4.1 ‘Categorical Syllogism’

Two categorical statements if logically formulated, taken together would lead to a conclusion. The two statements should serve as the premises. If there is no premises, then there will be a fallacy or enthymeme.

For example, the following two examples stand for categorical syllogisms:

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|---------------------------------|------------|----------------|
| I) All As are B | } | → Major Premis |
| | } Premises | |
| C is an A | } | → Minor Premis |
| Therefore, C is B | | → Conclusion |
| | | |
| II) All Politicians are corrupt | } | → Major Premis |
| | } Premises | |
| Blogg is a politician | } | → Minor Premis |
| Therefore, Blogg is corrupt | | → Conclusion |

The following is an example of a fallacy of enthymeme because it leaves out the premises:

Blogg is a politician.

Therefore, he is corrupt.

3.4.2 Hypothetical Syllogism

Hypothetical Syllogism is referred to as **Modes Poneus**, affirming the antecedent. The major premise has if.....then form.

The following are two examples:

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|--|---|-----------------------------|
| If A is true,
(Antecedent) | then B is true
(Consequent) | Therefore, B is true. |
| | | |
| If Blogg is a politician,
(Antecedent) | then he is corrupt
(Consequent) | Therefore, Blogg is corrupt |

Hypothesis statements may have a major Hypothesis (H) and auxiliary ($A_1 A_2 \dots A_n$) hypotheses.

3.4.3 ‘Logical Truth’ and ‘Material Truth’

All statements which are logically true are not necessarily “materially true”. For example:

All politicians have two tongues.

Blogg is a politician

Therefore, Blogg has two tongues.

The above statement is “logically true” but “materially” not true.

3.4.4 Deductive Fallacies

Deductive fallacies are those where premises do not lead necessarily to the stated conclusions. There are three main types of deductive fallacies:

- 1) Logical (formal) Fallacies (Fallacy of affirming the consequent)

Correct Reasoning or ‘Modus Poneus’ (Affirming the antecedent)	‘Logical (Formal) Fallacy’ (Affirming the consequent)
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If A is true, then B is true.
A is true.
Therefore, B is true.

If A is true, B is true.
B is true.
Therefore, A is true.

2) Verbal Fallacies (Fallacy of Composition)

The verbal fallacy involves a statement where something which is true of the part is also made true of the whole. For example, in case of people waiting to see a procession:

If one rises on one's tip-toes, one can see better.

If people rise on their tip-toes, they can see better.

3) Material Fallacies (All others): Material fallacies are of several kinds.

a) **Post hoc ergo propter hoc:** (After this, therefore because of this).

Because event B occurs after event A, then event B is necessarily caused by event A.

b) "Argument by Analogy": It refers to a statement where, "A is similar to B, therefore whatever is true of A is also true of B". For example, Singapore was backward in 1950s. It developed faster because of "open economy policies". India was backward in 1950s. Therefore, India would have prospered by 'open economy policies'.

c) "Appeal to Authority": It refers to a statement where the truth is sought to be asserted by referring to an authority. "A is true because (so & so) say it so". For example, "India is shining because Milton Friedman has said so".

Check Your Progress 1

1) What do you mean by the term 'Scientific explanation'?

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2) What is the difference between Phenomenalism and Nominalism.

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3) State the rules of logic. Explain 'categorical syllogism'.

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4) What is deductive fallacy?

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3.5 MODELS OF SCIENTIFIC EXPLANATION

“Men do not think they know a thing till they have grasped the ‘why’ of it”

- Aristotle

In the light of understanding of the basic building blocks of knowledge viz., facts, testing and formal logic, we shall now turn to the Positivist Models of Scientific Explanation. The basic building blocks of scientific explanation are theories. The status, structure and functions of theories and theoretical terms become essential part of scientific explanation. The aim of scientific research is not only to discover and describe events and phenomena in the world but also, and more importantly, to explain and understand **why** these phenomena occur as they do.

The early Positivists like Aguste Comte and Ernest Mach did not believe in any role to explanation in science. Mach believed that theories are mere heuristic devises and talked of theories being “eliminative fiction” useful only in organizing data. But Positivism has traveled a long distance since, and by mid-1950s was referring to explanation as the Chief objective of Science. “To explain the phenomena in the world of our experience, to answer the question “Why?” rather than only the question “What?” is one of the foremost objectives of all rational enquiry...”. Explanation removes puzzlement and provides understanding. However, there exist considerable differences of opinion as to the function and the essential characteristics of scientific explanation.

What follows is the description of some of the basic models of scientific explanation within the positivist approach. First, we shall discuss the hypothetico-Deductive Model (H-D Model), and then turn to Covering Law Models encompassing the Deductive-Nomological (D-N) and Inductive-Probablistic (I-P) Models.

3.5.1 Hypothetico-Deductive Model

At a time when the Positivists were not ready to concede any role to theoretical explanation, **Carnap and Hempel**, in their writings, came out with a model – which later came to be known as **Hypothetico-Deductive (H-D) model** of explanation. The H-D model not only describes the structure of theories, but provides answers to the questions of the status and functions of theories, as well. H-D model explicitly addresses the problems of a theory’s structure. The propositions in a deductive system are seen as being arranged in an order of levels. Higher-level hypotheses will refer to theoretical entities, all which need not be tested. Lower-level hypotheses describe observable phenomena and are tested against reality. H-D model addresses the problem of status of theoretical

terms which may not be testable directly, to gain meaningfulness indirectly by the successful confirmation of theory in which they are embedded.

In H-D model, theories as a whole are tested by comparing their deduced consequences (prediction) with data. Every theoretical term need not be given empirical counterparts via correspondence rules. H-D model also turns the old **realist-instrumentalist controversy** into a moot debate. **Realists** claim that all theoretical terms must refer to real entities and theories, which do not, are false. Instrumentalists insist that theories are only instruments for predictions. Only relevant question for them is whether theories are adequate for prediction. H-D model seems to accommodate both these concerns and defuse the controversy.

The H-D model rejects the earlier notion that theories have no role but are only instruments. H-D model emphasizes the following positive functions of theories:

- 1) They allow generality in the specification of scientific laws.
- 2) They possess 'a certain formal simplicity' which allows the use of 'powerful and elegant mathematical machinery'.
- 3) They can serve the practical function of allowing the scientist to discover interdependencies among observable.
- 4) They are convenient and fruitful heuristic (intellectual) devices, often serving an explanatory function of their own.

Thus, the H-D model without minimizing the importance of observable phenomena for scientific explanation, allowed far more substantial role for theories and theoretical terms than did their predecessors.

3.5.2 Covering-Law Models

Carl G. Hempel and **Paul Oppenheim** developed Deductive-Nomological (D-N) Model of scientific explanation and much later, Hempel extended it to include Inductive-Probabilistic (I-P) Model. These two models together are referred to as Covering-Law Models of explanation.

3.5.2.1 Deductive-Nomological (D-N) Model

Hempel and Oppenheim in their paper entitled "Studies in the Logic of Explanation" advanced an account of scientific explanation, which later came to be known as deductive-nomological (D-N) model. They divided an explanation into two major constituents – the **explanandum** and the **explanans**. The **Explanandum** means the sentence describing the phenomenon to be explained (not that phenomenon itself). **The Explanans** mean the class of those sentences, which are adduced to account for the phenomenon. The explanans fall into two sub-classes, viz., certain antecedent conditions $C_1, C_2, \dots, C_k$ and certain general laws $L_1, L_2, \dots, L_3$. If a proposed explanation is to be sound, conditions of adequacy must be satisfied. The following four conditions of logical and empirical adequacy must be satisfied:

i) **Logical Conditions of Adequacy**

- R_1) The explanandum must be a logical consequence of the explanans; in other words, the explanandum must be logically deduced from the

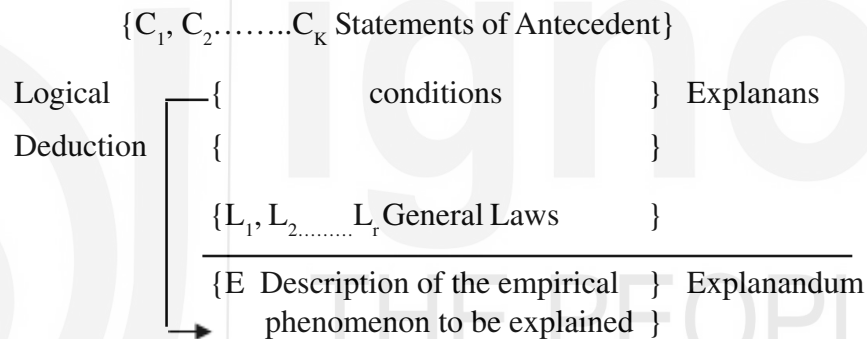
information contained in the explanans, for otherwise, the explanans would not constitute adequate grounds for the explanandum.

- R₂) The explanandum must contain general laws, and these must actually be required for the derivation of the explanandum. Though not a necessary condition, the explanans must contain at least one statement which is not a law.
- R₃) The explanans must have empirical content, i.e., it must be capable, at least in principle, of test by experiment or observation. The point deserves special mention because certain arguments which have been offered as explanations in the natural and in the social sciences violate this requirement.

ii) Empirical Condition of Adequacy

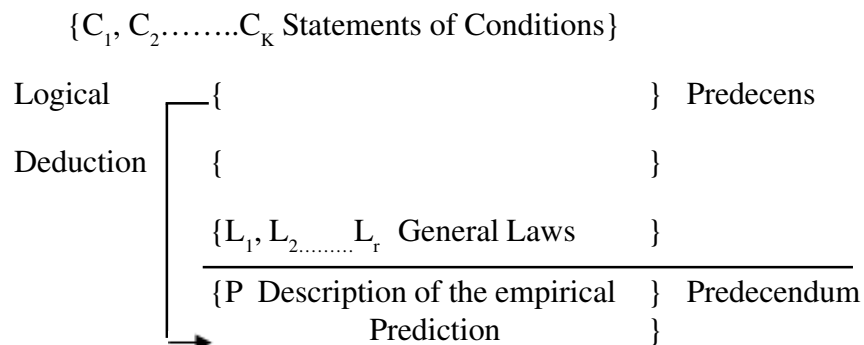
- R₄) The sentences constituting the explanans must be true.

The characteristics of explanation discussed may be summarized into the following schema:



Hempel and Oppenheim argued that the same formal analysis, including the four necessary conditions, applies to scientific predictions as well as to explanation. They went on to argue that an explanation is not fully adequate unless its explanans, if taken account of in time, could have served as a basis for predicting the phenomenon under consideration. Consequently, whatever is said about the logical characteristics of explanation or prediction will be applicable to either, even if one of them is mentioned. This resulted what is known as Covering-Law Model extended to the D-N Model of Explanation.

Extension of characteristics of explanation to prediction could formally be presented in the following schema:



The symmetry visualized between explanation and prediction has often drawn criticism.

3.5.2.2 Inductive-Probabilistic (I-P) Model

In 1960s Hempel developed Inductive-Probabilistic Model. In the I-P model the explanans consists of sentences describing the requisite initial conditions along with statistical laws and is expected to confer upon the explanandum highly logical, or inductive probability. The characteristics are similar to D-N Model. The symmetry holds and hence another Covering-Law Explanation.

Check Your Progress 2

- 1) Identify the basic blocks of knowledge.

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- 2) Explain how Hypothetico-Deductive Model defuses the conflict between the realists and instrumentalists.

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- 3) State two characteristics of Deductive-Nomological (D-N) Model.

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3.5.2.3 Critical Appraisal of Covering-Law (D-N, I-P) Models of Explanation

The Covering-Law Model of explanation offered by the D-N and I-P models are subject to a number of limitations. For convenience we shall confine here to the limitations of D-N model, some of which are also applicable to I-P model.

First is about the necessity of laws and theories in the pursuit of scientific explanation. Laws are distinct from ordinary statements. Ordinary statements or statement of mere facts refer to particular statements and express “contingent truths”, like “John went home and had dinner”. Laws refer to universal statements and express “necessary truths”. For example, “all objects are attracted to one another with a force inversely proportional to the square of the distances between them” is a universal and law like statement: However, there are many apparently

universal statements which are not really universal statements. These may be the ones that involve mere “accidental generalizations” like, “all coins in my pocket are nickels”.

This distinction between laws and ordinary statements does not make laws as necessary requirement for explanation. Particularly in biology, astronomy and social sciences explanations need not be in the form of laws. Explanations may be in the form of questions and answers. The often cited example is: Why do Hopi Indians continue to perform rain making ceremonies even though their ceremonies do not normally produce desired result? The explanation is to reinforce group identity, but there is no law involved. Similarly: Why do all vertebrates’ hearts beat regularly? The explanation is to circulate blood throughout the body which helps to maintain uniform temperature. This explanation again doesn’t require any law like statement.

Second, explanation requires not only the factors one cites to be casually relevant but also that they be causes. D-N model is often satisfied by citing relevant causes. Causes explain effects, but effects do not explain their causes, and effects of common cause do not explain one another. Based on David Hume’s famous guillotine of time, two contiguous events do not become cause and effect. Inferring event B as the effect of event A because it followed the later, amounts to the “Billiard Ball Model of Causation”, where the ball reaching the pouch is seen, ‘as if’ the player knew all the laws of time and motion of the billiard ball.

Third, much of the criticism of Covering Law Model of explanation relates to the symmetry thesis, that explanation and prediction are inherently linked, and that only difference is the sequence. But critics point out that explanation need not predict. For example, Darwin’s Theory of Evolution provides a great deal of explanation on the origin and evolution of species, but it does not predict anything. Similarly, prediction need not explain anything. For instance, prediction of weather does not explain anything.

Fourth, the D-N Model ignores the divergence between the norms and practices and thereby eliminates much of science from the ambit of explanation. Many functional or technological explanations get eliminated because they are not through any laws or regularities. Purposive behaviour needs explanation of motivation and in motivated behaviour future affects the present. Determining motives to be classified in antecedent conditions and not in causal relation. Hempel and Oppenheim also recognize the problems in applying their model of explanation in social sciences. Human behaviour often is characterized by uniqueness and irrepeatability. It depends upon situation and previous history of individuals.

Notwithstanding all these limitations, “many philosophers would still maintain that the D-N Model is an important starting point for studying scientific explanation and there is something right and important about it.” (Hausman, 1994, p.9). Particularly in the mainstream economics, scientific explanation is strongly associated with Positivist models of explanation and thus D-N Model becomes an important reference point for the debates on explanation in economics.